



Executive Summary

Total Rentals

6,172,314 trips

Average Daily Rentals

16,910 /day

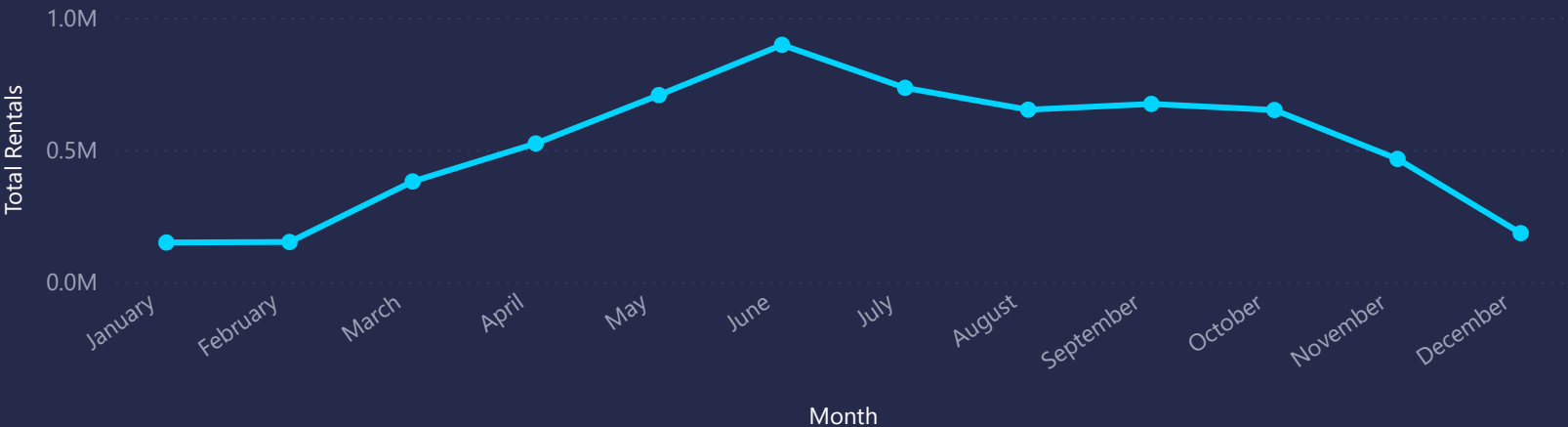
Peak Hour

18:00

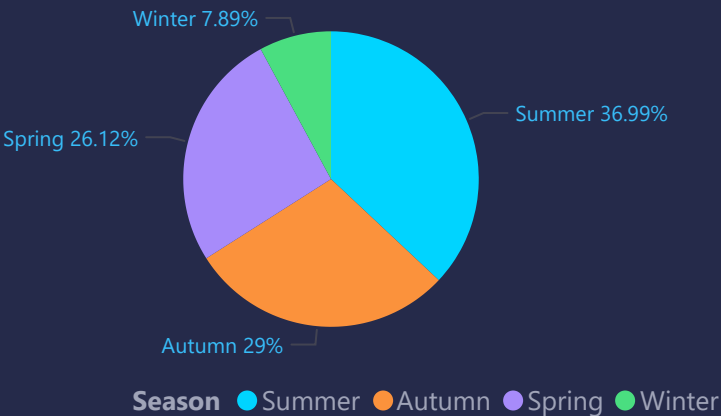
Bussiest Season

Summer

Total Rentals by Month



Total Rentals by Season



full_date	day_name	season_name	day_type	total_rentals	avg_temperature
Friday, January 26, 2018	Friday	Winter	Working Day	2931	-14.74
Wednesday, January 24, 2018	Wednesday	Winter	Working Day	3154	-14.08
Thursday, January 25, 2018	Thursday	Winter	Working Day	3113	-14.00
Tuesday, January 23, 2018	Tuesday	Winter	Working Day	3981	-10.92
Thursday, January 11, 2018	Thursday	Winter	Working Day	4017	-10.77
Friday, January 12, 2018	Friday	Winter	Working Day	4111	-10.70
Saturday, January 27, 2018	Saturday	Winter	Working Day	2693	-10.43
Tuesday, February 06, 2018	Tuesday	Winter	Working Day	3988	-10.01
Sunday, February 04, 2018	Sunday	Winter	Working Day	2487	-9.52
Tuesday, December 12, 2017	Tuesday	Winter	Working Day	5496	-9.47
Monday, February 05, 2018	Monday	Winter	Working Day	3932	-9.40
Wednesday, December 13, 2017	Wednesday	Winter	Working Day	6019	-8.62

Temporal Analysis

Weekday Average

719.4 /hr

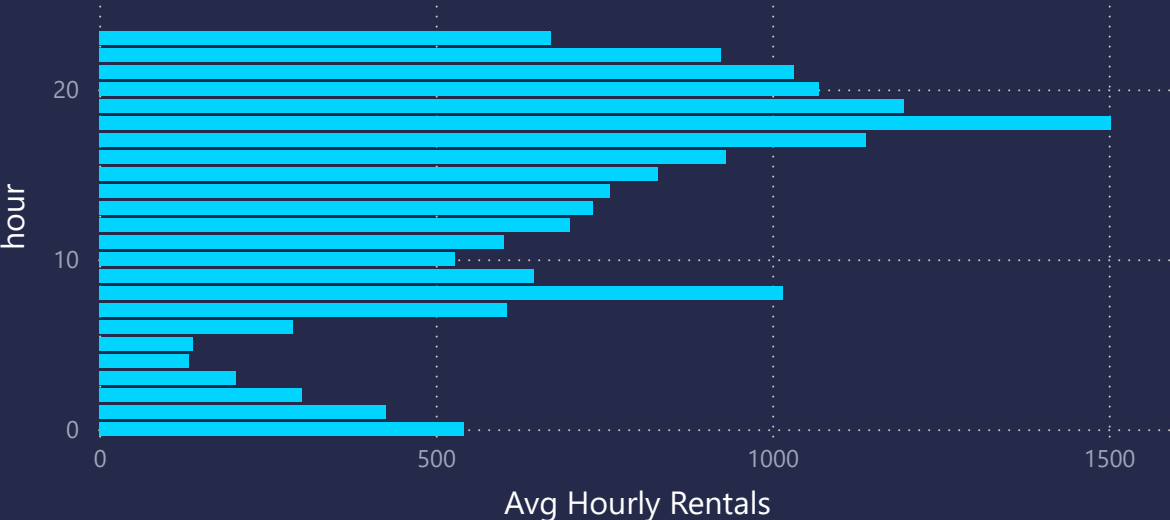
Weekend Average

667.3 /hr

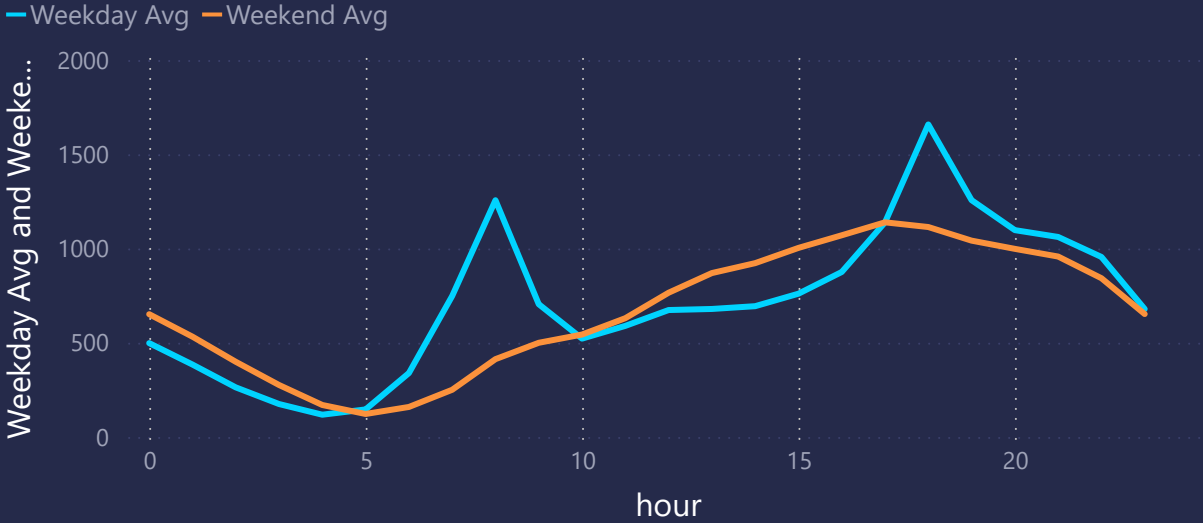
Rush Hour Percentage

36.1%

Avg Hourly Rentals by hour



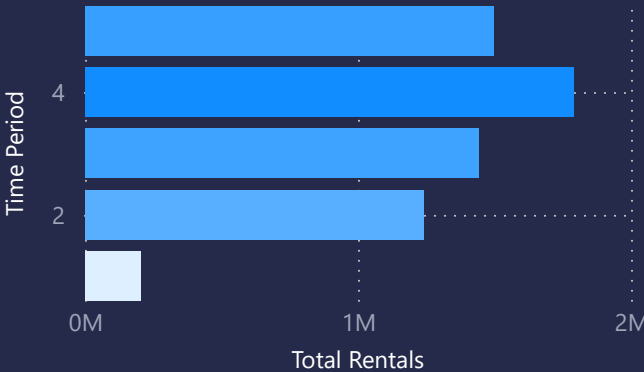
Weekday Avg and Weekend Avg by hour



Heatmap : Day x Hour

day_name	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Monday	460	296	200	133	103	157	370	797	1269	704	522	595	680	689	713	783	908	1173	1695	1294	11
Tuesday	473	380	260	171	115	141	335	755	1276	715	519	573	654	663	668	702	817	1055	1552	1166	10
Wednesday	506	396	277	192	127	142	321	727	1250	723	563	632	717	725	741	807	921	1195	1668	1278	11
Thursday	520	415	289	184	119	142	321	705	1197	671	479	553	641	635	647	708	803	1048	1605	1216	10
Friday	529	432	287	193	126	148	346	754	1287	708	525	593	675	682	701	799	925	1220	1767	1326	11
Saturday	664	597	439	300	185	139	187	295	487	567	595	679	794	906	959	1047	1116	1209	1159	1063	10
Sunday	639	467	358	251	154	105	131	205	338	432	491	582	736	833	885	959	1024	1069	1069	1020	9
Total	541	426	302	203	133	139	288	606	1016	646	528	601	699	733	759	829	931	1139	1503	1195	10

Total Rentals by Time Period



Weather Impact

Average Temperature

12.9 °C

Rain Impact Percentage

↓ 78.2% vs Clear

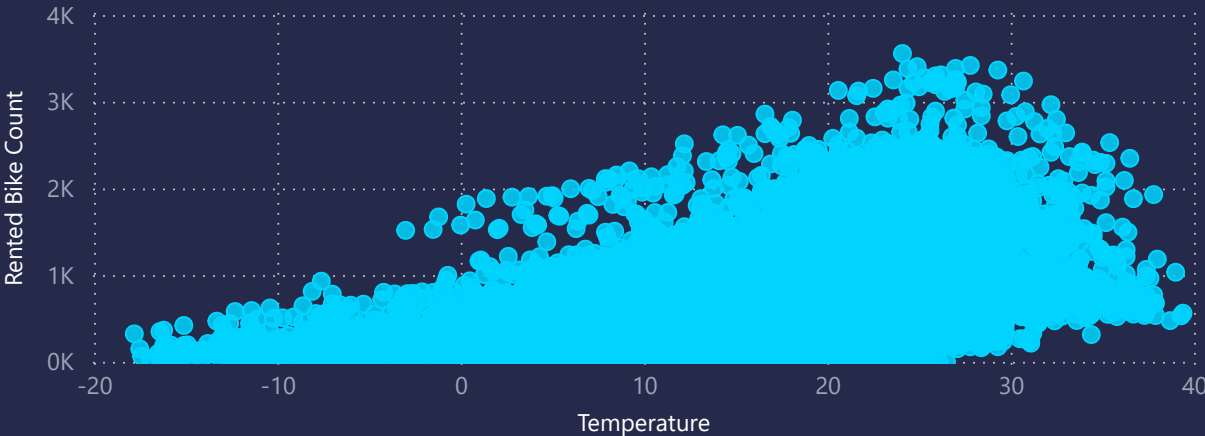
Snow Impact Percentage

↓ 75.1% vs Clear

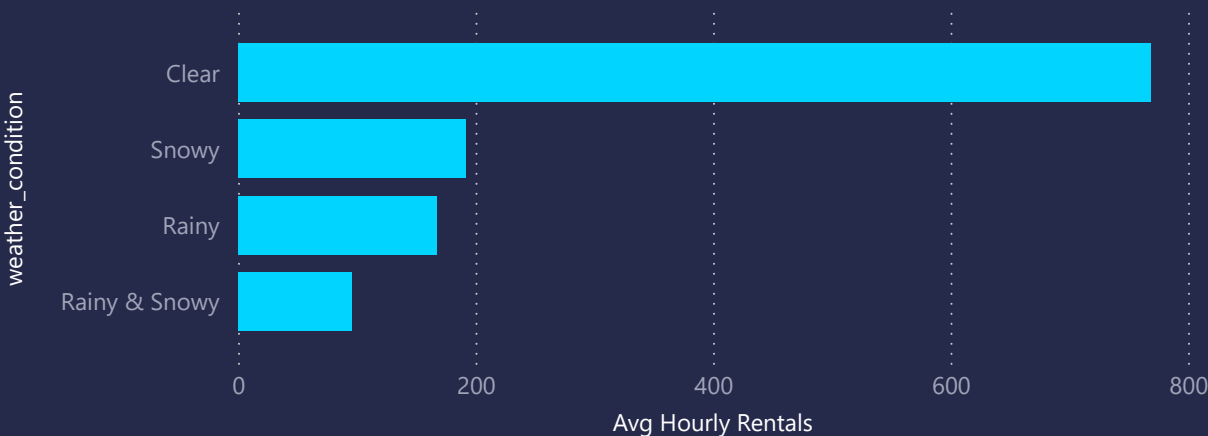
Optimal Temperature Range

Hot

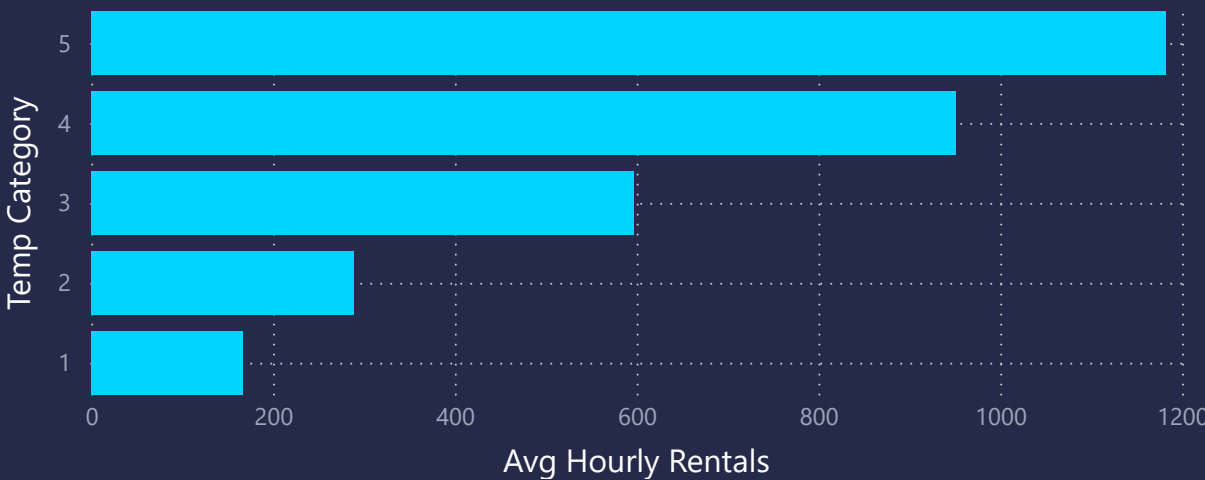
Temperature and Rented Bike Count



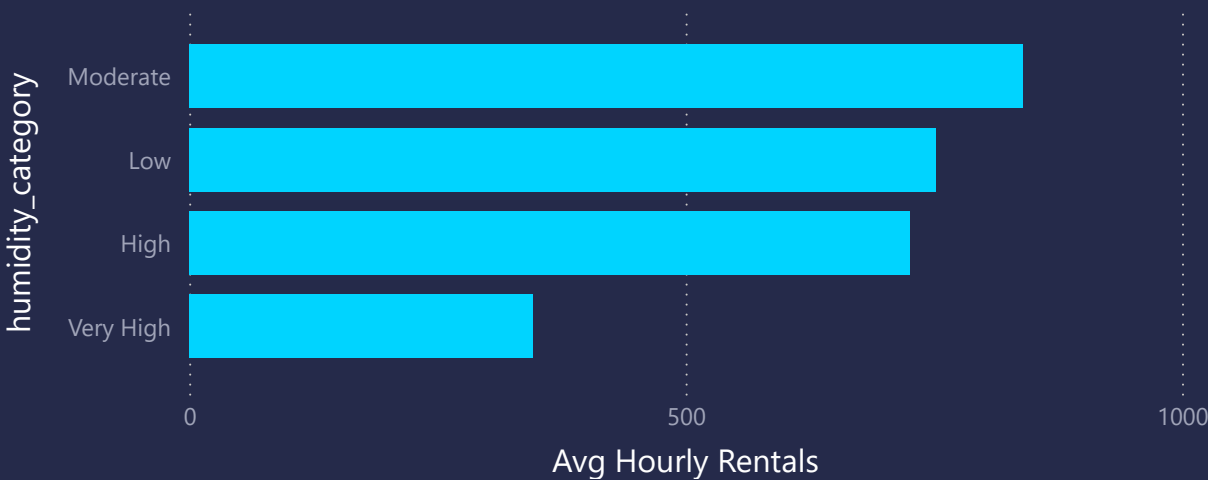
Avg Hourly Rentals by Weather Condition



Avg Hourly Rentals by Temp Category



Avg Hourly Rentals by Humidity



Operational Intelligence

Zero Demand Hours

295 hours (3.4%)

Demand Volatility

0.915 (Moderate)

Demand StDev

$\sigma = 645.0$

Holiday Impact Factor

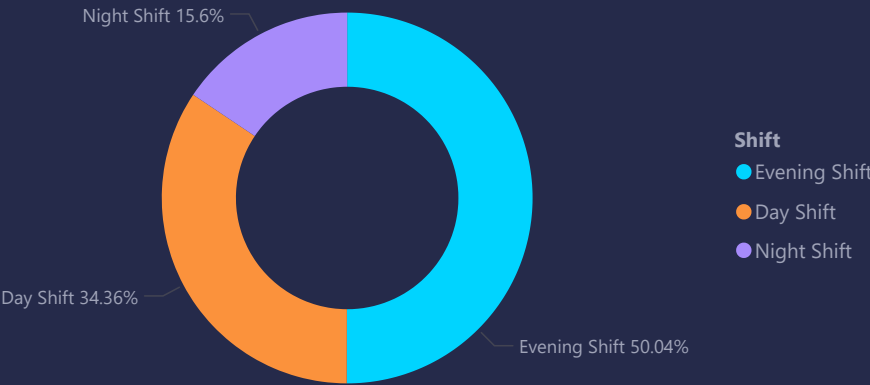
0.72x — ↓ Lower

shift	Holiday	Non-Functioning	Working Day	Total
Day Shift	592	0	759	726
Evening Shift	739	0	1112	1058
Night Shift	257	0	346	330
Total	529	0	739	705

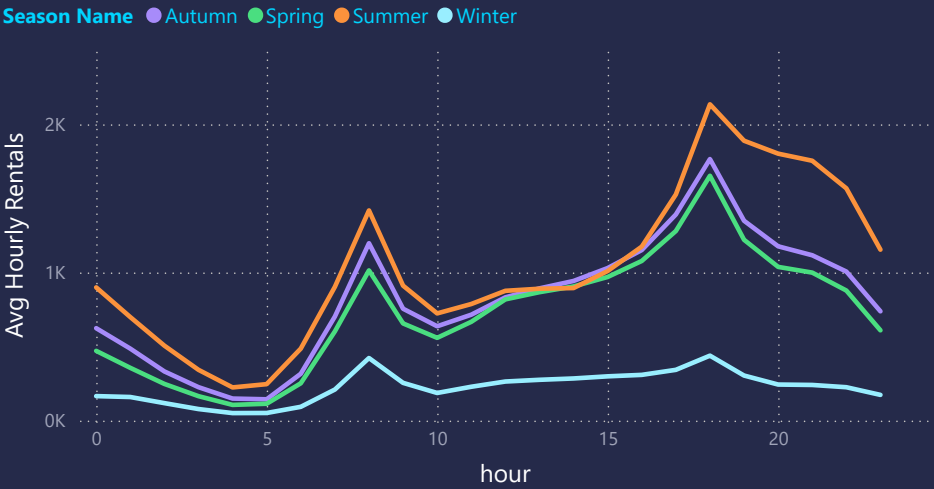
Avg Hourly Rentals by Day Type



Total Rentals by Shift



Avg Hourly Rentals by Hour and Season



Business Recomendation

Season Name

Autumn

Spring

Summer

Winter

Weather Condition

Clear

Rainy

Rainy & Snowy

Snowy

Day Category

Weekday

Weekend

Estimated Revenue

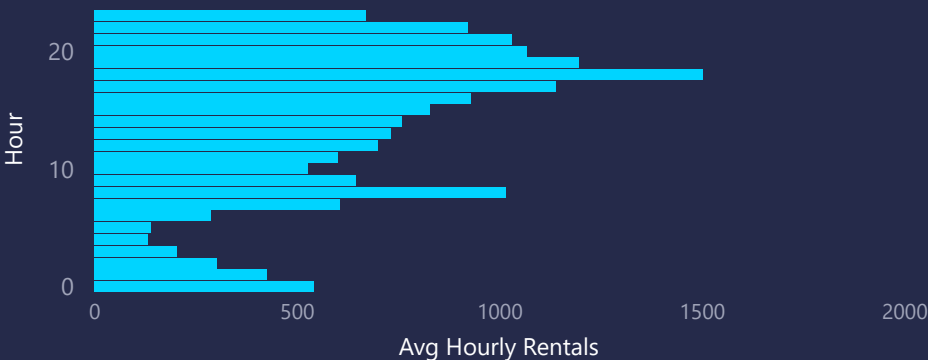
₩9,258.5M

Revenue Lost During Rain

₩450.9M lost

hour	Avg Hourly Rentals	Action Recommendation
0	541	Moderate - Maintain
1	426	Moderate - Maintain
2	302	Moderate - Maintain
3	203	Low - Promote/Discount
4	133	Low - Promote/Discount
5	139	Low - Promote/Discount
6	288	Low - Promote/Discount
7	606	Moderate - Maintain
8	1016	High - Maximize Supply
9	646	Moderate - Maintain
10	528	Moderate - Maintain
11	601	Moderate - Maintain
12	699	Moderate - Maintain
13	733	High - Maximize Supply
14	759	High - Maximize Supply
15	829	High - Maximize Supply
16	931	High - Maximize Supply
17	1139	High - Maximize Supply
18	1503	High - Maximize Supply
19	1195	High - Maximize Supply
20	1069	High - Maximize Supply

Avg Hourly Rentals by Hour



Estimated Revenue by Season

